

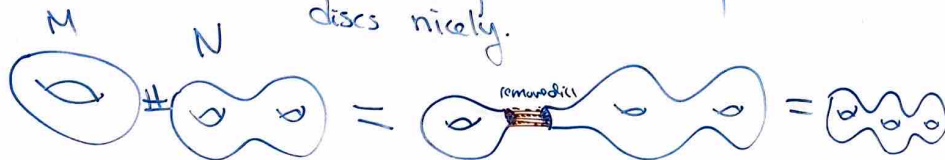
Building 2-Manifolds

(Reminder: 2-mfd $\rightarrow$ locally homeomorphic to $\mathbb{R}^2$)

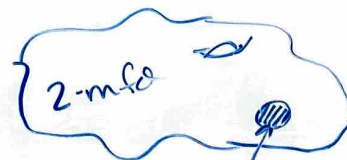
M, N are 2-mfds

Connected sum

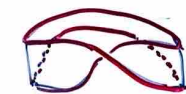
$M \# N$ = removing a disc from each and gluing along the boundary of the missing discs nicely.



add cross-cap



remove a disc
glue in a
Möbius band



$$\partial M \equiv S^1 \equiv \partial B_2$$

what we're adding in

what we removed

these homeomorphisms composition is another homeomorphism, the gluing map.

add a handle



remove 2 discs
add in a tube/handle
by gluing ∂ to the
2 holes "nicely"
Using a homeomorphism

Classification of Compact 2-manifolds (w/out boundary)

Every cpt 2-manifold is
homeomorphic to one of the
following:

$$\left[\begin{array}{l} S, \mathbb{T}^2, \mathbb{T}^2 \# \mathbb{T}^2, \overbrace{\mathbb{T}^2 \# \mathbb{T}^2 \# \mathbb{T}^2}^{3\text{-fold torus}}, \dots \\ \mathbb{P}^2, \mathbb{P}^2 \# \mathbb{P}^2, \dots \end{array} \right] \begin{array}{l} \text{orientable} \\ \text{non-orientable} \end{array}$$

2-fold projective plane

Questions

- ① What about both?
- ② What about a circle?
- ③ What about a line?

Classification of Compact
2-manifolds (w/out boundary)

Every 2-manifold is
homeomorphic to one of the

$\pi^2 \# \pi^2, \pi^2 \# \pi^2 \# \pi^2, \dots$ 3-fold torus
 $\mathbb{P}^2 \# \mathbb{P}^2, \dots$ 2-fold projective plane

Orientable
 non-orientable

Questions:

- ① Where is the Klein bottle in this?
- ② What happens if we add a cross-cap to π^2 or $\mathbb{P}^2$?
- ③ What happens if we add a handle to π^2 or $\mathbb{P}^2$?

CW complex

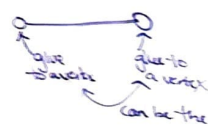
Can rep. topo spaces
(not just manifolds)

X = the topo space we're building

• 0-cells = vertices = X_0

• 1-cells = edges

= homeomorphic to B_1



$$\partial B^1 \equiv S^0$$

X_1 = set of edges

1-skeleton
(graph)

• 2-cells $\cong B^2$

$\partial B^2 = S^1$ ← can make w/
a sequence of edges

X_2 = set of 2-cells

$$X_k = \text{cells}; \bigcup_{k \in \mathbb{N}} X_k = K\text{-skeleton}$$

2-skeleton

X is a topo space,
The Euler characteristic is

$$\chi(X) := |X_0| - |X_1| + |X_2| - \dots$$

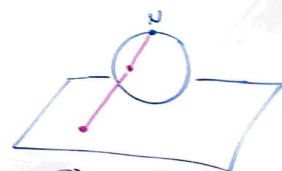
Q What is χ of a
cpt 2-manifold?

$$\chi(S^2) := 2$$

plane graphs:

$$M - |E| + |F| = 2$$

2 regions

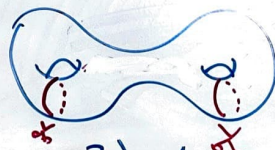


$$S^1 \setminus u \cong \mathbb{R}^1$$

S^2 is the 1-point
compactification of $\mathbb{R}^2$

The genus of a surface is
the number of ^{disjoint} copies of S^1
that can be removed w/out
disconnecting the surface

usually denoted
w/ g .



$$g(\pi^2) = 1$$

$$g(\pi^2 \# \pi^2) = 2$$

$$g(\underbrace{\# \pi^2}_{k\text{-times}}) = k$$

$$g(S^2) = 0$$

$$g(\mathbb{P}^2) = 1$$

$$g(K \cong \mathbb{P}^2 \# \mathbb{P}^2) = 2$$

Questions

- ① When
bottu
- ② What
a cro
- ③ What
a ha

Theorem

$M = 2\text{-}m^{\text{cpt}}\text{fld}$ of genus g

$$\chi(M) = \begin{cases} 2-2g, & \text{orientable} \\ 2-g, & \text{non-orientable} \end{cases}$$

"Doubling": Non-orientable to orientable!

- think of non-orientable surface as being "thick".

- new surface: "the outside" of the thick surface

